SIDANG TUGAS AKHIR —
TAHUN AJARAN
2025/2026

Company Profile

Di Universitas Pembangunan Nasional "Veteran" Jakarta | Kayra Sahadewa (0099160298) | Pembimbing: Desi Anita, S.Kom | PPL SMK Tunas Media

SMK Tunas Media - UPN "Veteran" Jakarta

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About the Company

- UPN "Veteran" Jakarta memiliki 8 laboratorium komputer aktif (LAB-RPL, MULTIMEDIA, JARINGAN, IOT, dll) dengan intensitas pemakaian tinggi lebih dari 120 jam per minggu.
- Pengelolaan jadwal konvensional via papan tulis dan ketiadaan display otomatis



Layar Display Kiosk TV 8 Laboratorium Full HD di Lorong Kampus

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Our Mission and Vision

Landasan utama dalam perancangan dan pengembangan sistem informasi display jadwal lab.

Solusi Usulan Terpadu

Membangun sistem berbasis web dengan WebSocket Socket.IO zero-reload, fitur permohonan pinjam lab mandiri, dan live check-in praktikum.

Rumusan Masalah

Write the company's vision dan sinkronisasi display seketika (< 500 ms) di lorong laboratorium?

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Our Goals

Tiga sasaran strategis utama pengembangan sistem display laboratorium komputer.

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1. Otomasi Validasi Jadwal

Menjamin 100% bebas bentrok (conflict-free).



2. Display Kiosk Real-Time

Menampilkan status okupansi 8 lab di layar TV lorong 24/7 dengan transmisi WebSocket (< 500 ms).



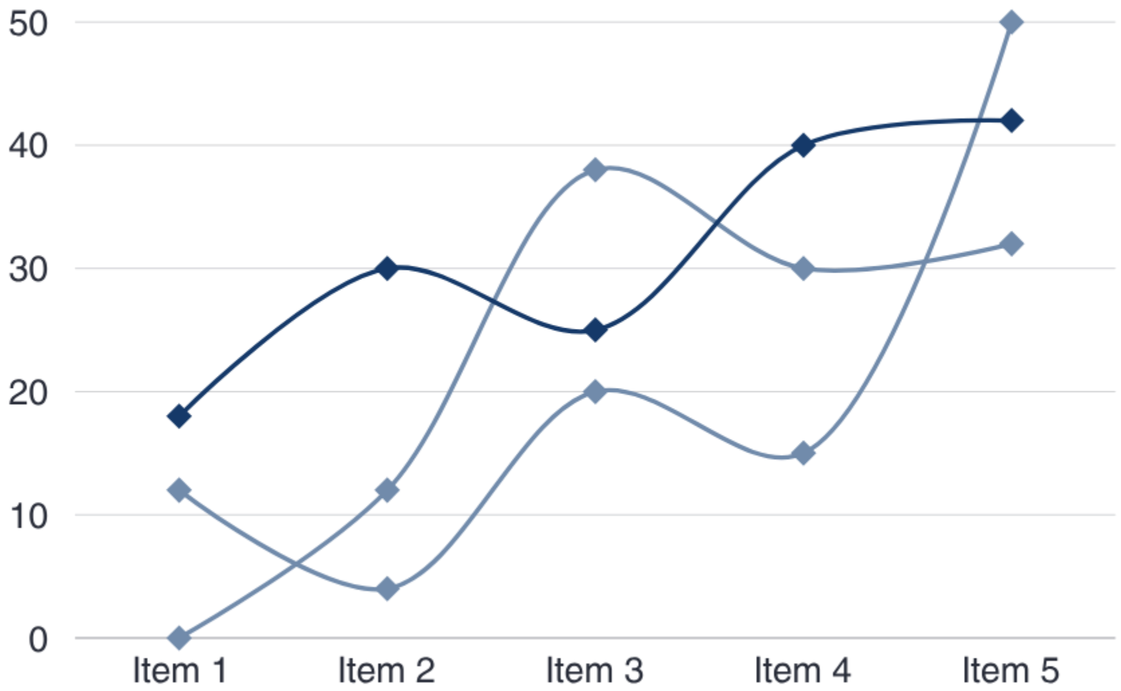
3. Digitalisasi Administrasi

Digitalisasi alur peminjaman lab (proposal 5MB), check-in praktikum, pelaporan PC error, dan ekspor Excel.

Indikator Operasional & Capaian Sistem

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Metrik operasional 8 laboratorium komputer aktif di UPN "Veteran" Jakarta.



Laboratorium Komputer Terintegrasi ke Display TV

8 LAB

Total Intensitas Jam Perkuliahan & Praktikum per Minggu

> 120 JAM

Latensi Transmisi Data Real-Time WebSocket Zero-Reload

< 500 MS

Our Market

Perbandingan efisiensi alur kerja operasional lama dibandingkan sistem usulan.

SISTEM LAMA

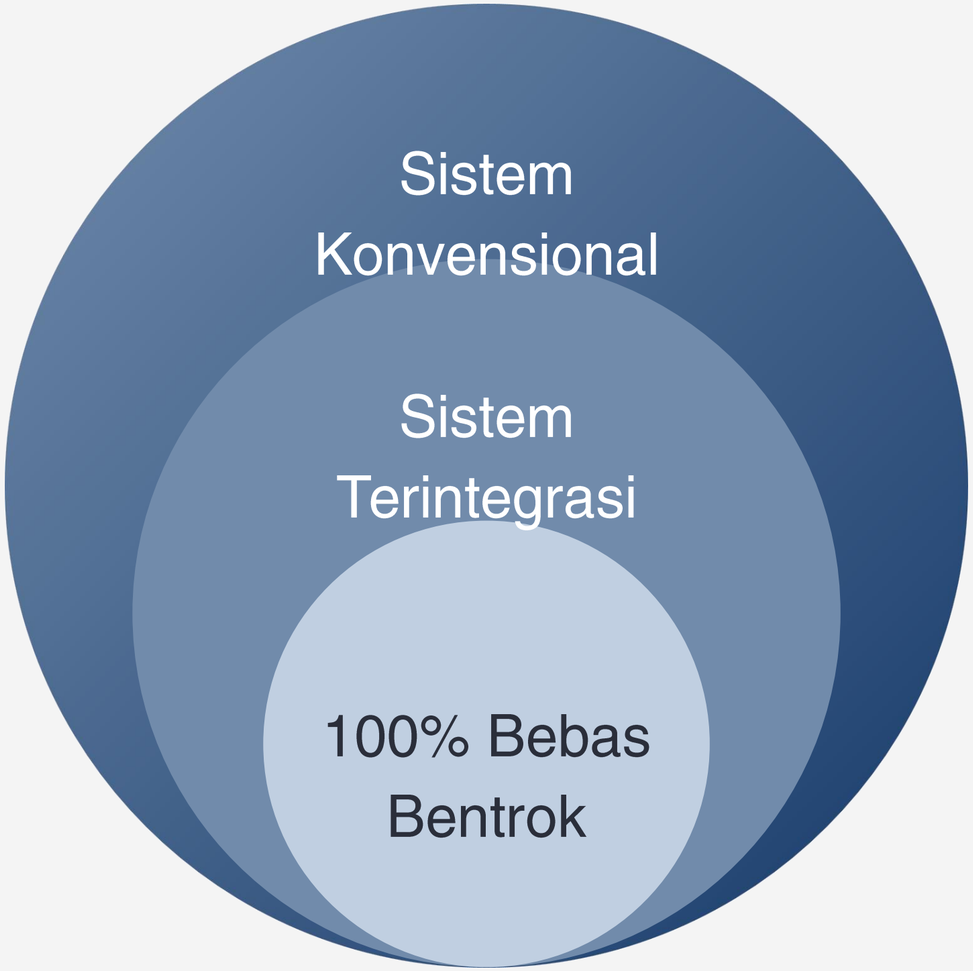
Total Available izin pinjam lab dengan formulir fisik berbelit.

SISTEM USULAN

Basis data terpusat 24/7, pemblokiran bentrok otomatis seketika, layar Kiosk Full Live,

HASIL EFISIENSI

Eliminasi 100% bentrok jadwal dan transparansi status ruang bagi seluruh civitas akademika UPNV J



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Our Milestones

Pendekatan bertahap sekuensial guna menjamin mutu, stabilitas, dan akurasi perangkat lunak.

01. Analisis

Wawancara dengan Laboran dan observasi langsung pada 8 lab di UPNVJ.

02. Desain

Perancangan Use Case, ERD basis data, dan arsitektur WebSocket Gateway.

03. Pengkodean

Implementasi Frontend Vue 3 TailwindCSS dan Backend Node.js / Express.

04. Pengujian

Pengujian Whitebox (Basis Path overlap) dan Blackbox (12 skenario uji klien).

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Hak Akses Sistem (RBAC Matrix)

Pemisahan wewenang aktor secara tegas demi integritas data dan keamanan operasional.

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Administrator

Kelola master 8 lab, manajemen akun & role, kalender fakultas, siaran darurat.



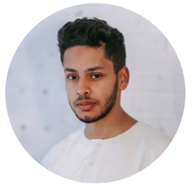
Laboran

Verifikasi jadwal kuliah, approval peminjaman lab, sesi praktikum, rekap Excel.



Dosen & Mahasiswa

Melihat kalender lab, pengajuan pinjam + upload proposal 5MB, lapor PC rusak.



Display Kiosk TV

Layar pasif 24/7 di koridor kampus, update seketika via WebSocket tanpa login.



Sistem Daemon

Sinkronisasi otomatis status jadwal dan validasi background.

Our Offered Services

Modul fungsional terintegrasi untuk mendukung seluruh kebutuhan operasional laboratorium.

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- 1. Kalender Jadwal Interaktif & Filter Lab Fleksibel
- 2. Engine Deteksi Bentrok Otomatis (Conflict-Free)
- 3. Pengajuan Peminjaman Mandiri & Upload Proposal 5MB
- 4. Live Check-In / Check-Out Sesi Praktikum Mahasiswa
- 5. Pelaporan Tiket Kendala PC Rusak & Ekspor Excel

Financial Guidance

Evaluasi mutu perangkat lunak berstandar akademik Bab IV Tugas Akhir.

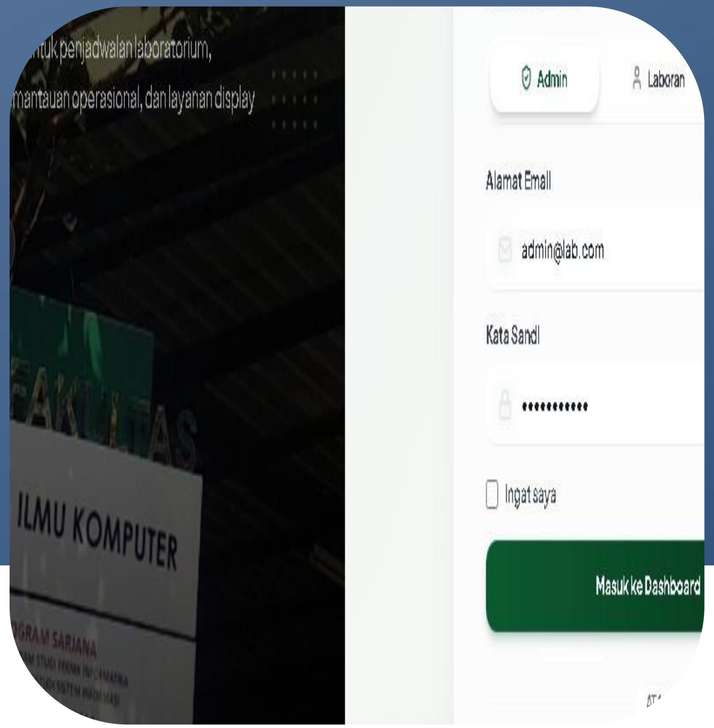
*Pengujian menyeluruh logika internal dan fungsionalitas antarmuka sistem	White Box Testing	Black Box Testing	Hasil Evaluasi
Cakupan Uji	100% LULUS	100% LULUS	%
Metrik Pengujian			
Kecepatan Respon			
Status Kelaikan			

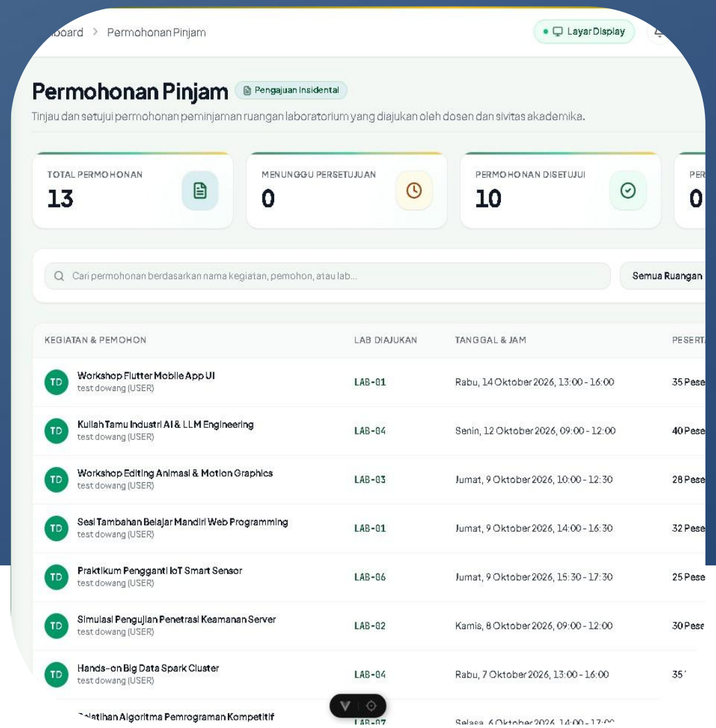
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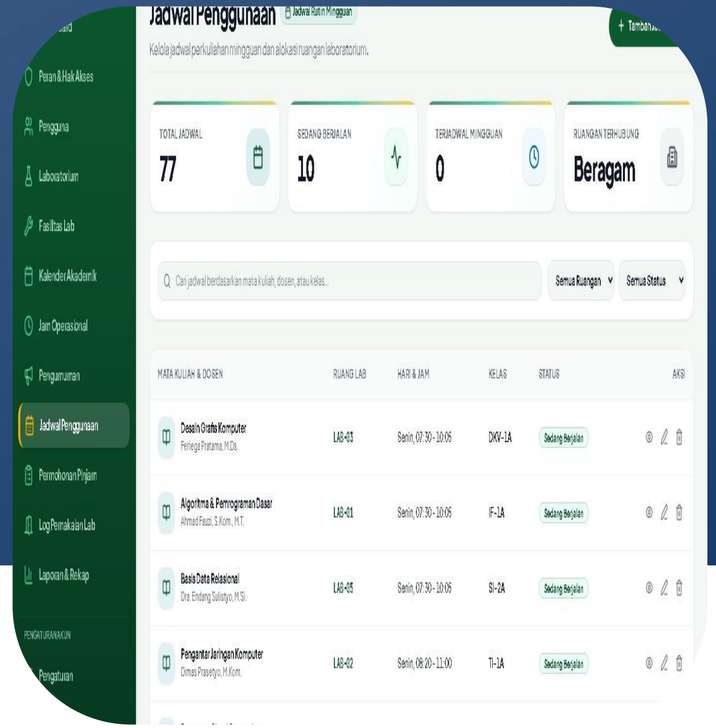
To merge, **first highlight** two or more cells, then click **"Merge Cells"** to organize your table according to your needs!

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Implementasi Antarmuka: Display Kiosk TV & Dasbor Manajemen Jadwal







Layar TV Kiosk lorong 1080p menampilkan status live 8 lab secara seketika (< 500 ms) tanpa reload browser.

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Connect with us.

Program Studi Rekayasa Perangkat Lunak

Kayra Sanadewa (0099160298)

SMK Tunas Media

Guru Pembimbing: Desi Anita, S.Kom.

Sesi Tanya Jawab (Q&A) & Uji Petik

Universitas Pembangunan Nasional "Veteran" Jakarta



Collaborate on a Whiteboard









Write a note here



Write a note here

Tip: Collaboration makes teamwork easier! Click "Share" and invite your teammates to fill this up. Use this page for bulletins, brainstorms, and other fun team ideas.

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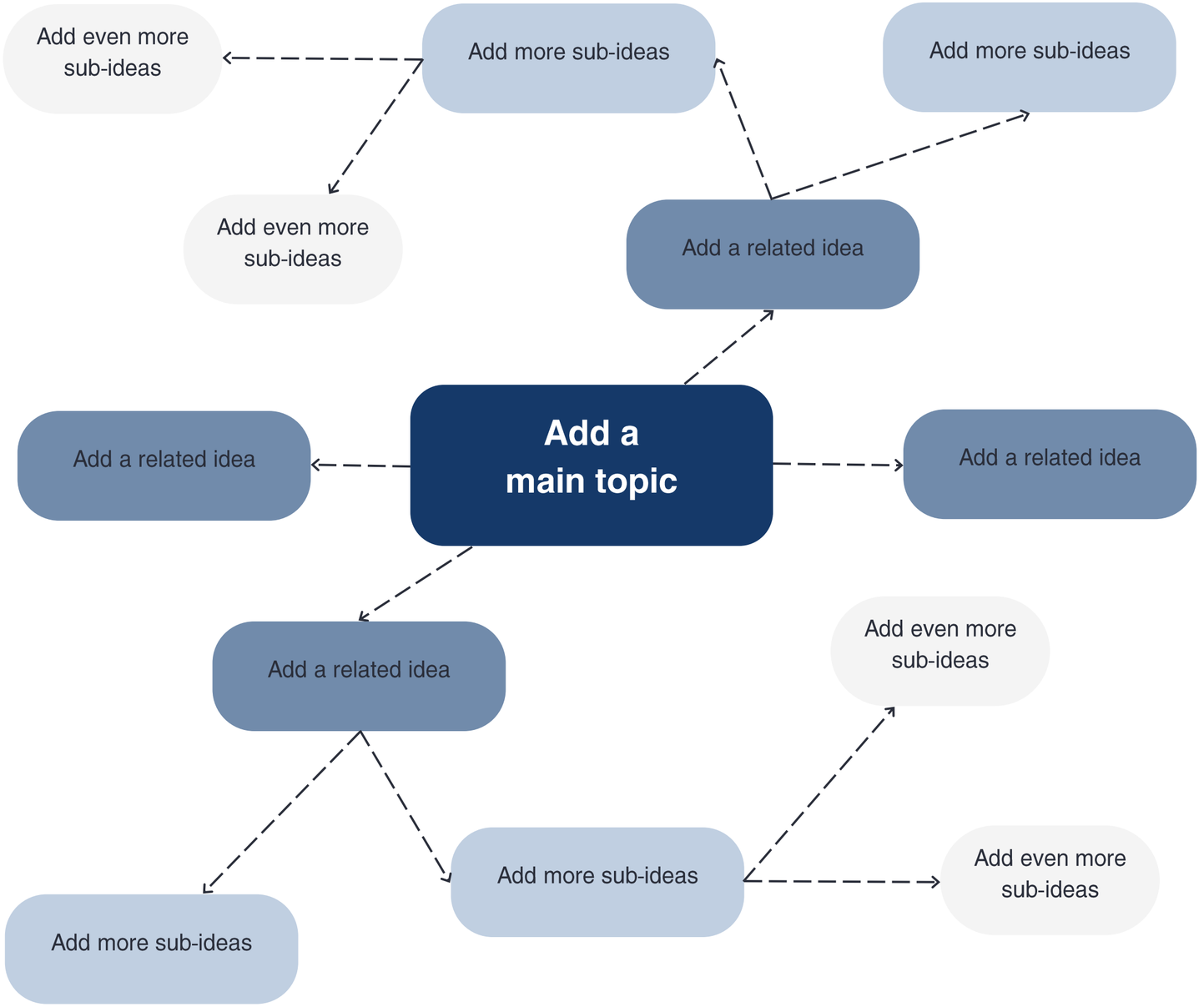
Copy a note, drag to the board, and write your ideas.

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